Heterogeneous CSI

Widar
- Complex CSI
- Gesture recognition

Gait
- Complex CSI
- User identification

XRF55
- Amplitude-oriented
- Activity recognition

ElderAL
- Amplitude-only
- Activity recognition

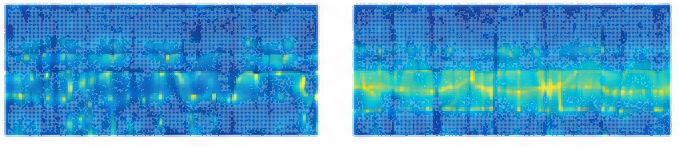
Complex vs. amplitude-only

Regular vs. irregular sampling

Different sampling rates

Different phase structure

Example CSI observations



Complex CSI | Amplitude-only

Configurable preprocessing benchmark

Unified parsing & labels

Dataset loading

CSI format parsing

Label grouping & splitting

Configurable preprocessing pipeline

Complex CSI: Denoise → Outlier → Calibrate → Normalize → Interpolate

Amplitude-only: Denoise → Outlier → Phase skip → Normalize → Interpolate

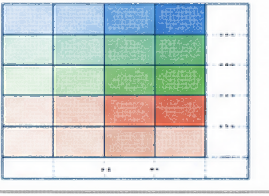
Stage I: Preset-model screening

6 preprocessing presets × 19 models

Preprocessing presets (6) × Learning models (19): MLP / CNN / ResNet / LSTM / Transformer / … → Model performance comparison

Stage II: Fine-grained evaluation

320 / 80 preprocessing–model combinations

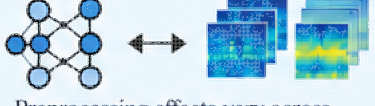


Model–preprocessing dependency analysis

Mechanism analysis and preprocessing guidelines

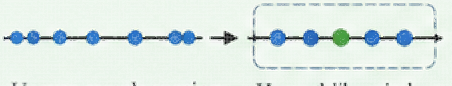
Key findings

1 Effects depend jointly on data and model.



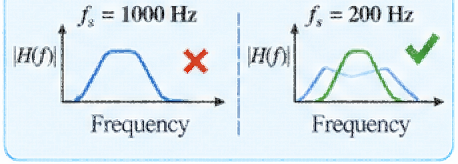
Preprocessing effects vary across datasets and models.

2 Fixed-frame locality can fail under irregular sampling.

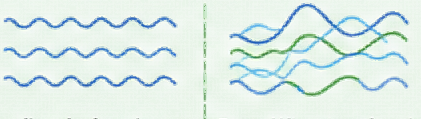


Uneven sample spacing — Hampel-like window

3 Filter effectiveness depends on sampling-aware parameterization and signal semantics.



4 Independent subcarrier detrending can disrupt phase structure.



Aligned subcarriers | Rotated/diverging subcarriers